

## 1 Gain law

(bars denote normalized quantities: lengths by  $R$ ;  $a_o := \Delta t / (\gamma_N R)$  [1/pN] and  $a_{\text{nom}} := \Delta t / \gamma_N = a_o R$  [ $\mu\text{m}/\text{pN}$ ];  $\bar{a} := a / a_{\text{nom}} = a / (a_o R)$  for the physical gain  $a$  [ $\mu\text{m}/\text{pN}$ ]; forces by  $a_o$ :  $\bar{f} := a_o f$ , so that  $\bar{a} \bar{f} = \Delta \bar{w}$  ( $\bar{f}_{dw}$  the applied force,  $\bar{f}_T$  the thermal force);  $0 \leq \bar{a} \leq 1$ , far field 1, contact 0; hats sit on the normalized quantities,  $\hat{a}_w, \hat{b}, \delta \hat{w}$ ;  $e_\theta = \theta - \hat{\theta}$ )

$$\bar{a}_{\text{true}}(\bar{w}) = 1 - \frac{9}{8\bar{w}} + \frac{1}{2\bar{w}^3} - \frac{57}{100\bar{w}^4} + \frac{1}{5\bar{w}^5} + \frac{7}{200\bar{w}^{11}} - \frac{1}{25\bar{w}^{12}}$$

$$\bar{a}'_{\text{true}}(\bar{w}) > 0 \quad \text{on} \quad \bar{w} > 1 \quad \implies \quad \bar{a}' = b_{\text{true}}(\bar{a})(1 - \bar{a})^2 \quad \text{exactly,} \quad b_{\text{true}} := \frac{\bar{a}'_{\text{true}}}{(1 - \bar{a}_{\text{true}})^2}$$

$$\begin{aligned} \bar{a}_{\text{true}}(1) = 0, \quad \bar{a}'_{\text{true}}(1) = 1 &\implies b_{\text{true}}(\bar{a} = 0) = 1 \\ 1 - \bar{a}_{\text{true}} = \frac{9}{8\bar{w}} + \mathcal{O}(\bar{w}^{-3}) &\implies b_{\text{true}}(\bar{a} \rightarrow 1) = \frac{8}{9} \end{aligned}$$

$$\begin{aligned} \bar{a}'(\bar{a}, b) &:= b(1 - \bar{a})^2 \\ a_w &= a_o \bar{a}_w, \quad a_o = \frac{\Delta t}{\gamma_N R} \end{aligned}$$

## 2 Jacobians

$$\begin{aligned} \frac{\partial \bar{a}'}{\partial \bar{a}} &= -2b(1 - \bar{a}) \\ \frac{\partial \bar{a}'}{\partial b} &= (1 - \bar{a})^2 = \frac{\bar{a}'}{b} \end{aligned}$$

## 3 True dynamics

$$(\delta \bar{w}_j[k] := \delta \bar{w}[k - d + j - 1], \quad j = 1, \dots, d + 1, \quad d = 2)$$

$$\begin{aligned} \bar{w}[k] &= \bar{w}_d[k] - \delta \bar{w}_3[k] \\ \bar{w}_d[k + 1] &= \bar{w}_d[k] + \Delta \bar{w}_d[k] \\ \delta \bar{w}_1[k + 1] &= \delta \bar{w}_2[k] \\ \delta \bar{w}_2[k + 1] &= \delta \bar{w}_3[k] \\ \delta \bar{w}_3[k + 1] &= \lambda_c \delta \bar{w}_3[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k] \\ \bar{a}_w[k + 1] &= \bar{a}_w[k] + \bar{a}'_{\text{true}}[k] \left[ \Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \right] \\ b[k + 1] &= b[k] \end{aligned}$$

$$F_{dw}[k] = \bar{f}_{dw}[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{f}_{dw}[k - i]$$

$$\varepsilon_w[k] = \bar{a}_w[k] \bar{f}_T[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{a}_w[k - i] \bar{f}_T[k - i] - (1 - \lambda_c) n_w[k]$$

## 4 Process model

$$\mathbf{x} = [\delta\bar{w}_1, \delta\bar{w}_2, \delta\bar{w}_3, \bar{a}_w, b]^\top \quad (d = 2)$$

## 5 Estimator

$$\delta\hat{w}_1[k + 1] = \delta\hat{w}_2[k]$$

$$\delta\hat{w}_2[k + 1] = \delta\hat{w}_3[k]$$

$$\delta\hat{w}_3[k + 1] = \lambda_c \delta\hat{w}_3[k]$$

$$\hat{a}_w[k + 1] = \hat{a}_w[k] + \hat{b}[k] (1 - \hat{a}_w[k])^2 [\Delta\bar{w}_d[k] + (1 - \lambda_c) \delta\hat{w}_3[k]]$$

$$\hat{b}[k + 1] = \hat{b}[k]$$

## 6 Error dynamics

(True – Estimator; coefficients at the estimate)

$$e_{\delta\bar{w}_1}[k + 1] = e_{\delta\bar{w}_2}[k]$$

$$e_{\delta\bar{w}_2}[k + 1] = e_{\delta\bar{w}_3}[k]$$

$$e_{\delta\bar{w}_3}[k + 1] = \lambda_c e_{\delta\bar{w}_3}[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k]$$

$$\begin{aligned} e_{\bar{a}_w}[k + 1] = & \left[ 1 + \hat{b}(1 - \hat{a}_w)^2 F_{dw} - 2\hat{b}(1 - \hat{a}_w)(\Delta\bar{w}_d + (1 - \lambda_c)\delta\hat{w}_3) \right] e_{\bar{a}_w}[k] \\ & + (1 - \lambda_c) \hat{b}(1 - \hat{a}_w)^2 e_{\delta\bar{w}_3}[k] + (1 - \hat{a}_w)^2 (\Delta\bar{w}_d + (1 - \lambda_c)\delta\hat{w}_3) e_b[k] \\ & + \hat{b}(1 - \hat{a}_w)^2 \varepsilon_w[k] \end{aligned}$$

$$e_b[k + 1] = e_b[k]$$

$$F_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dw} & 0 \\ 0 & 0 & (1 - \lambda_c) \hat{b}(1 - \hat{a}_w)^2 & 1 + \hat{b}(1 - \hat{a}_w)^2 F_{dw} & (1 - \hat{a}_w)^2 \\ 0 & 0 & 0 & -2\hat{b}(1 - \hat{a}_w)(\Delta\bar{w}_d + (1 - \lambda_c)\delta\hat{w}_3) & \times (\Delta\bar{w}_d + (1 - \lambda_c)\delta\hat{w}_3) \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{q}[k] = \begin{bmatrix} 0, & 0, & -\varepsilon_w, & \hat{b}(1 - \hat{a}_w)^2 \varepsilon_w, & 0 \end{bmatrix}^\top$$

## 7 Measurement and innovation

$$\begin{aligned} \nabla_d \bar{w}_d[k] &= \bar{w}_d[k] - \bar{w}_d[k-d] \\ y_1[k] &= \delta \bar{w}_1[k] + n_w[k] \\ y_2[k] &= \bar{a}_w[k-d] + n_a[k] \end{aligned}$$

( $n_a$ : readout noise on the normalized gain;  $\text{Var} = R_2$ . The filter is fed the whitened increment of this readout, EWMA weight  $a_{\text{cov}}$ , and the second row of  $H$  carries the same factor.)

$$\bar{a}_w[k-d] = \bar{a}_w[k] - b(1 - \bar{a}_w[k])^2 \nabla_d \bar{w}_d[k]$$

$$\begin{aligned} \frac{\partial y_2}{\partial \bar{a}_w} &= 1 + 2\hat{b}(1 - \hat{a}_w) \nabla_d \bar{w}_d \\ \frac{\partial y_2}{\partial b} &= -(1 - \hat{a}_w)^2 \nabla_d \bar{w}_d \end{aligned}$$

$$H[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 + 2\hat{b}(1 - \hat{a}_w) \nabla_d \bar{w}_d & -(1 - \hat{a}_w)^2 \nabla_d \bar{w}_d \end{bmatrix}$$

$$L[k] = P^f H^\top (H P^f H^\top + R)^{-1}$$

$$\begin{aligned} e_{y_1}[k] &= e_{\delta \bar{w}_1}[k] + n_w[k] \\ e_{y_2}[k] &= \left[ 1 + 2\hat{b}(1 - \hat{a}_w) \nabla_d \bar{w}_d \right] e_{\bar{a}_w}[k] - (1 - \hat{a}_w)^2 \nabla_d \bar{w}_d e_b[k] + n_a[k] \end{aligned}$$

## 8 Process and measurement noise

$$\begin{aligned} \kappa_T &:= \frac{4k_B T a_o}{R} = \frac{4k_B T a_{\text{nom}}}{R^2} \\ \text{Var}(\bar{a}_w[k] \bar{f}_T[k]) &= \kappa_T \bar{a}_w[k] \rightarrow \kappa_T \hat{a}_w[k] \\ \sigma_{n_w}^2 &= \frac{\sigma_n^2}{R^2} \end{aligned}$$

$$\begin{aligned}
Q_{33} &= \text{Var}(\varepsilon_w)[k] = \kappa_T \left( \hat{a}_w[k] + (1 - \lambda_c)^2 (\hat{a}_w[k-1] + \hat{a}_w[k-2]) \right) + (1 - \lambda_c)^2 \sigma_{n_w}^2 \\
Q_{34} = Q_{43} &= -\hat{b} (1 - \hat{a}_w)^2 Q_{33} \\
Q_{44} &= \hat{b}^2 (1 - \hat{a}_w)^4 Q_{33} \\
Q_{55} &= 0
\end{aligned}$$

$$\begin{aligned}
R &= \text{diag}(R_1, R_2) \\
R_1 &= \sigma_{n_w}^2 \\
R_2 &= 2a_{\text{cov}}^2 IF_{\text{var}}(\hat{a}_w) (\hat{a}_w + \xi)^2 \\
IF_{\text{var}}(\hat{a}_w) &= 1 + 2 \left[ c_1 u^2 + 2c_2 u(1 - u) + c_3 (1 - u)^2 \right], \quad u = \frac{\hat{a}_w}{\hat{a}_w + \xi} \\
\xi &= \rho_n \frac{\sigma_{n_w}^2}{\kappa_T}
\end{aligned}$$

( $u$  is the thermal share of the readout variance, so  $IF_{\text{var}}$  is re-evaluated at each step from  $\hat{a}_w$ .  $\rho_n$  is how much of the sensor noise the readout chain passes relative to the thermal drive, so  $\xi$  is that noise expressed as an equivalent  $\bar{a}$ .  $\rho_n$  is a pure ratio;  $\xi$  is in units of  $\bar{a}$ . Offline numbers:  $c_1 = 1.42044$ ,  $c_2 = 0.0098975$ ,  $c_3 = 0.098910$ ,  $\rho_n = 0.350941$ ,  $a_{\text{cov}} = 0.05$ .)

